

Volume 2 – Software Requirements Specification (SRS)

AI Quant Trading Platform

Version: 1.0

Status: Draft

Document Type: Software Requirements Specification

1. Introduction

Purpose

This document defines the complete software requirements for the AI Quant Trading Platform. It describes all functional modules, business logic, workflows, system behavior, interfaces, security requirements, and technical constraints required to build a production-grade multi-user SaaS platform.

2. Product Overview

The platform continuously scans cryptocurrency markets, evaluates trading opportunities using technical analysis and AI-assisted scoring, applies configurable risk management rules, and executes trades automatically through supported exchange APIs.

The architecture must support:

- Multi-user SaaS
 - Multi-exchange connectivity
 - Automated trading
 - AI-assisted decision support
 - High availability
 - Horizontal scalability
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3. System Modules

Module 1 – Authentication Service

Features

- User Registration
- Email Verification
- Login
- Logout
- Password Reset
- Change Password
- Two-Factor Authentication
- Session Management
- Device Management
- JWT Authentication
- Refresh Tokens
- API Authentication

Permissions

- User
 - Premium User
 - Admin
 - Super Admin
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Module 2 – User Profile

Features

- Personal Information
 - Profile Picture
 - Time Zone
 - Preferred Currency
 - Language
 - Notification Preferences
 - Security Settings
 - Risk Profile
 - Default Strategy
 - Trading Preferences
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Module 3 – Exchange Integration

Supported Exchanges (Phase-wise)

- Mudrex
- Binance

- Bybit
- OKX
- KuCoin
- Gate.io
- MEXC
- Bitget

Features

- API Key Management
- Secret Encryption
- Connection Testing
- Permission Validation
- Account Balance Sync
- Position Sync
- Order Sync
- WebSocket Streaming
- Reconnection Logic

Module 4 – Market Data Engine

Responsibilities

- Live Price Feed
- OHLCV Data
- Order Book
- Trades Stream
- Funding Rate
- Open Interest
- Liquidation Feed
- Market Dominance
- Stablecoin Flow

Refresh Rate

- Tick Data
- 1 Second
- 5 Seconds
- 1 Minute
- Configurable

Module 5 – Market Scanner

The scanner continuously evaluates all supported trading pairs.

Filters

- Minimum Volume

- Liquidity
- Spread
- Volatility
- Trend Direction
- Volume Spike
- ATR Threshold
- RSI Conditions
- MACD Confirmation
- Market Structure
- BTC Correlation
- ETH Correlation
- News Filter

Output

- Candidate Coins
- Rejected Coins
- Confidence Score
- Risk Score

Module 6 – Technical Analysis Engine

Indicators

Trend

- EMA
- SMA
- VWAP
- SuperTrend
- Ichimoku

Momentum

- RSI
- MACD
- Stochastic RSI
- CCI
- ROC

Volume

- OBV
- CMF
- MFI
- Volume Profile

Volatility

- ATR

- Bollinger Bands
- Keltner Channel
- Donchian Channel

Market Structure

- Support
- Resistance
- Breakout
- Retest
- Trendline
- Supply Zone
- Demand Zone
- Order Blocks
- Liquidity Zones

Module 7 – AI Decision Engine

Responsibilities

- Score every trading pair
- Evaluate probability
- Estimate confidence
- Estimate risk
- Rank opportunities
- Recommend actions

Output

- Buy
- Sell
- Hold
- Ignore

Each recommendation must include an explanation showing which factors contributed to the score.

Module 8 – Strategy Engine

Strategies

- Trend Following
- Scalping
- Swing Trading
- Momentum
- Breakout
- Pullback
- Mean Reversion
- DCA
- Grid

- Custom Strategy

Strategy Configuration

- Entry Rules
 - Exit Rules
 - Stop Loss Rules
 - Take Profit Rules
 - Time Filters
 - Risk Filters
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Module 9 – Risk Management Engine

Features

- Maximum Risk Per Trade
- Maximum Daily Loss
- Maximum Weekly Loss
- Maximum Monthly Loss
- Position Sizing
- Dynamic Stop Loss
- Trailing Stop
- Break-even Logic
- Partial Profit Booking
- Consecutive Loss Protection
- Cooldown Period

Risk Profiles

- Conservative
 - Moderate
 - Aggressive
 - Custom
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Module 10 – Order Management

Supported Actions

- Market Order
- Limit Order
- Stop Order
- Stop Limit
- OCO (where supported)

Order Lifecycle

- Created
- Submitted

- Accepted
 - Filled
 - Partially Filled
 - Cancelled
 - Rejected
 - Expired
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Module 11 – Portfolio Management

Features

- Total Balance
 - Available Balance
 - Unrealized PnL
 - Realized PnL
 - Asset Allocation
 - Portfolio Performance
 - Diversification Analysis
 - Exposure Analysis
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Module 12 – Notifications

Channels

- Email
- Telegram
- WhatsApp
- Push Notifications
- Discord
- Slack

Events

- Trade Executed
 - Position Closed
 - Stop Loss Triggered
 - Take Profit Reached
 - API Failure
 - Exchange Offline
 - Daily Summary
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Module 13 – Reports & Analytics

Reports

- Daily

- Weekly
- Monthly
- Annual

Analytics

- Win Rate
 - Loss Rate
 - Average Profit
 - Average Loss
 - Profit Factor
 - Sharpe Ratio
 - Maximum Drawdown
 - Strategy Comparison
 - Trade Journal
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Module 14 – Paper Trading

Features

- Virtual Balance
 - Real Market Data
 - Strategy Testing
 - Performance Reports
 - Replay Mode
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Module 15 – Backtesting

Capabilities

- Historical Data Testing
 - Parameter Optimization
 - Walk-forward Testing
 - Monte Carlo Simulation
 - Strategy Comparison
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Module 16 – Admin Panel

Management

- Users
- Roles
- Permissions
- Subscriptions
- Exchanges
- Strategies

- System Settings
- Logs
- Notifications
- API Usage

Monitoring

- Active Users
- Running Strategies
- Queue Status
- Server Health
- Error Tracking

4. Non-Functional Requirements

Performance

- API response under 300 ms for standard requests
- WebSocket updates with minimal latency
- Background jobs processed asynchronously

Availability

- Target uptime: 99.9%

Scalability

- Horizontal scaling for stateless services
- Separate workers for scanning and order execution

Security

- Encrypted API credentials
- HTTPS only
- Role-based access control
- Audit logging
- Rate limiting
- Secure secret storage

Reliability

- Automatic retry for transient failures
- Graceful handling of exchange outages
- Recovery after service restart

5. Acceptance Criteria

The first production release shall allow a user to:

1. Register and secure an account.
 2. Connect a supported exchange.
 3. Configure a trading strategy.
 4. Set a risk profile.
 5. Run paper trading successfully.
 6. Enable live trading after confirmation.
 7. View live portfolio updates.
 8. Receive notifications for trade events.
 9. Review detailed analytics and reports.
 10. Disable automation instantly using an emergency stop.
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6. Out of Scope (Version 1)

- Stock trading
 - Forex trading
 - Options analytics
 - Social copy trading
 - NFT trading
 - DeFi protocol management
 - On-chain wallet execution
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7. Document Dependencies

The following design documents are required before implementation:

- Volume 3 – System Architecture
- Volume 4 – Database Design
- Volume 5 – REST & WebSocket API Specification
- Volume 6 – AI Decision Engine Design
- Volume 7 – UI/UX Specification
- Volume 8 – DevOps & Deployment Guide
- Volume 9 – Security Architecture
- Volume 10 – Testing Strategy